

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

NEWTON'S LAWS OF MOTION

LECTURE-4

PHYSICS

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Topic to be covered.

- 5m ~~#~~ ^{Revision} Centripetal acc & Centripetal Force $\rightarrow$ C-13
- 5m ~~#~~ Safe velocity on Friction Road $\checkmark$
- 5m ~~#~~ Safe velocity on Banked Road $\checkmark$
- ~~5m~~ ~~#~~ Safe velocity on Friction as well as Banked Road $\checkmark$

✓ WORK HARD

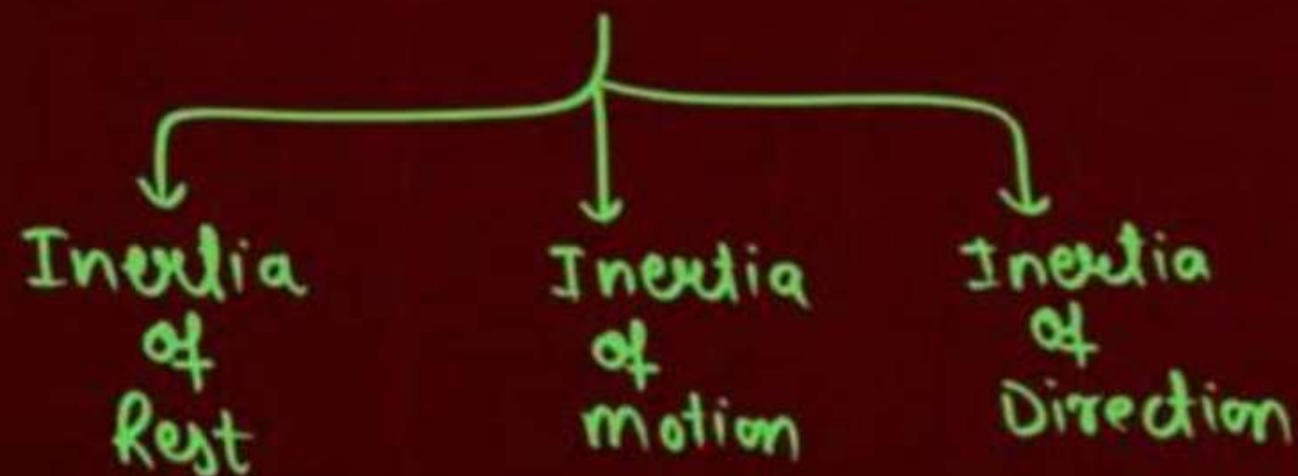
✓ DREAM BIG

✓ NEVER GIVE UP

Recap From 1st Lecture

Inertia (1st law of motion)

Property of Body by virtue of which Body cannot change its state of **Rest**, **motion** and **Direction**



Momentum (p)

$$\vec{p} = m\vec{v} \quad \text{kg m/s Vector}$$

$$\bullet p_f = mv \quad \bullet p_i = mv$$

② # Newton 2nd laws of Motion

$$F = \frac{\Delta p}{\Delta t} = ma \rightarrow \text{qth}$$

$$F = \frac{\Delta p}{\Delta t} = \frac{dp}{dt} = m \frac{dv}{dt} + v \frac{dm}{dt}$$

③ **Impulse**

Force acting for the short period of time.

$$\uparrow F = \frac{\Delta p}{\Delta t} \downarrow$$

$$F = \frac{dp}{dt} \quad \{F dt\} = dp$$

Impulse

Impulse momentum theorem

Recap From 2nd Lecture

Newton third law of motion

Action = -Reaction

- * Rocket propulsion
- * Walking
- * Boat
- * Recoiling on gun
- * Flying of Bird.

law of conservation of momentum

if $F_{ext} = 0$

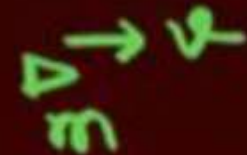
$$P_i = P_f$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

Application of law of conservation of momentum

① Recoiling velocity of gun

$$V = -\frac{mv}{M}$$



② Rocket Propulsion

$$F = m \frac{dv}{dt} + v \frac{dm}{dt}$$



? why Newton 2nd law is Max Real law of Motion

$$F = \frac{dP}{dt}$$

2nd law derive $V = u$

3rd law Derived
 $f_{12} = -f_{21}$

Lecture no. 3

Types of forces.

- * Gravitation force
- * Spring force
- * Tension
- * Normal Force
- * Electrostatic force
- * Friction force ✓

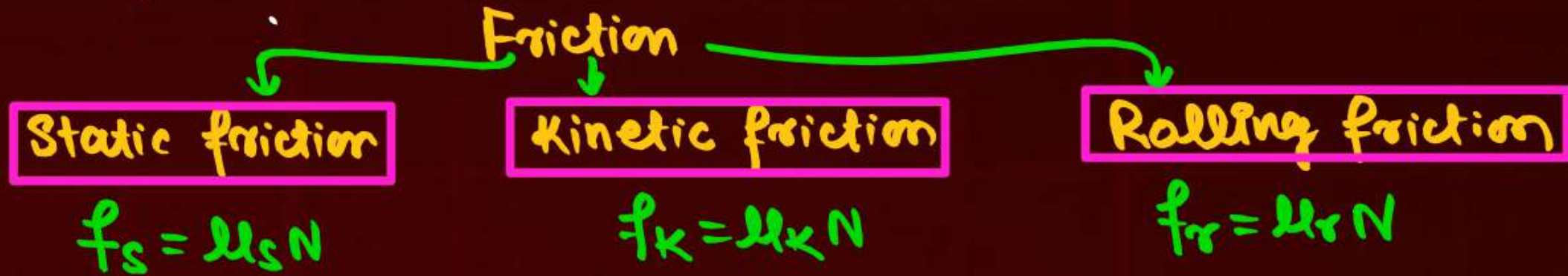
Friction

- * Force oppose the Relative Motion $f = \mu N$

- * Roughness & Adhesion

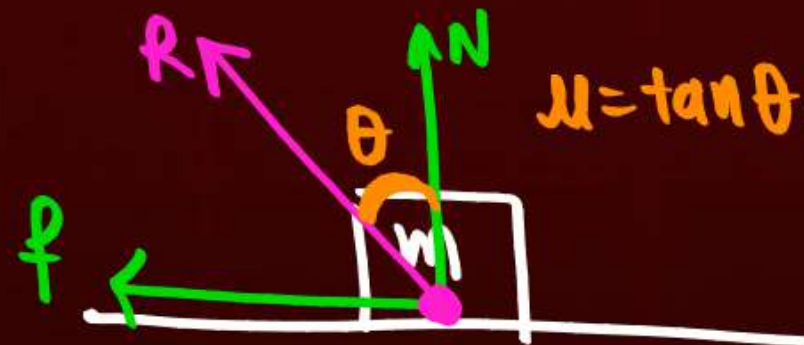
Laws of friction

- $f = \mu N$
- friction is independent of Area.

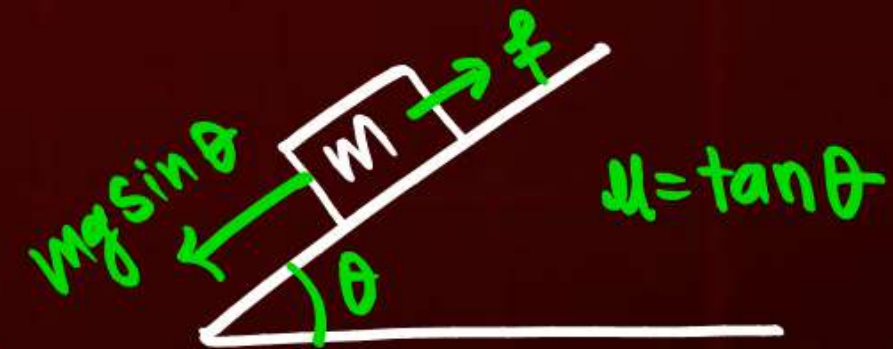


- ✓ Advantage of friction ✓
- ✓ Disadvantage of friction ✓
- ✓ How to Reduce the friction. ✓

Angle of friction



Angle of Repose



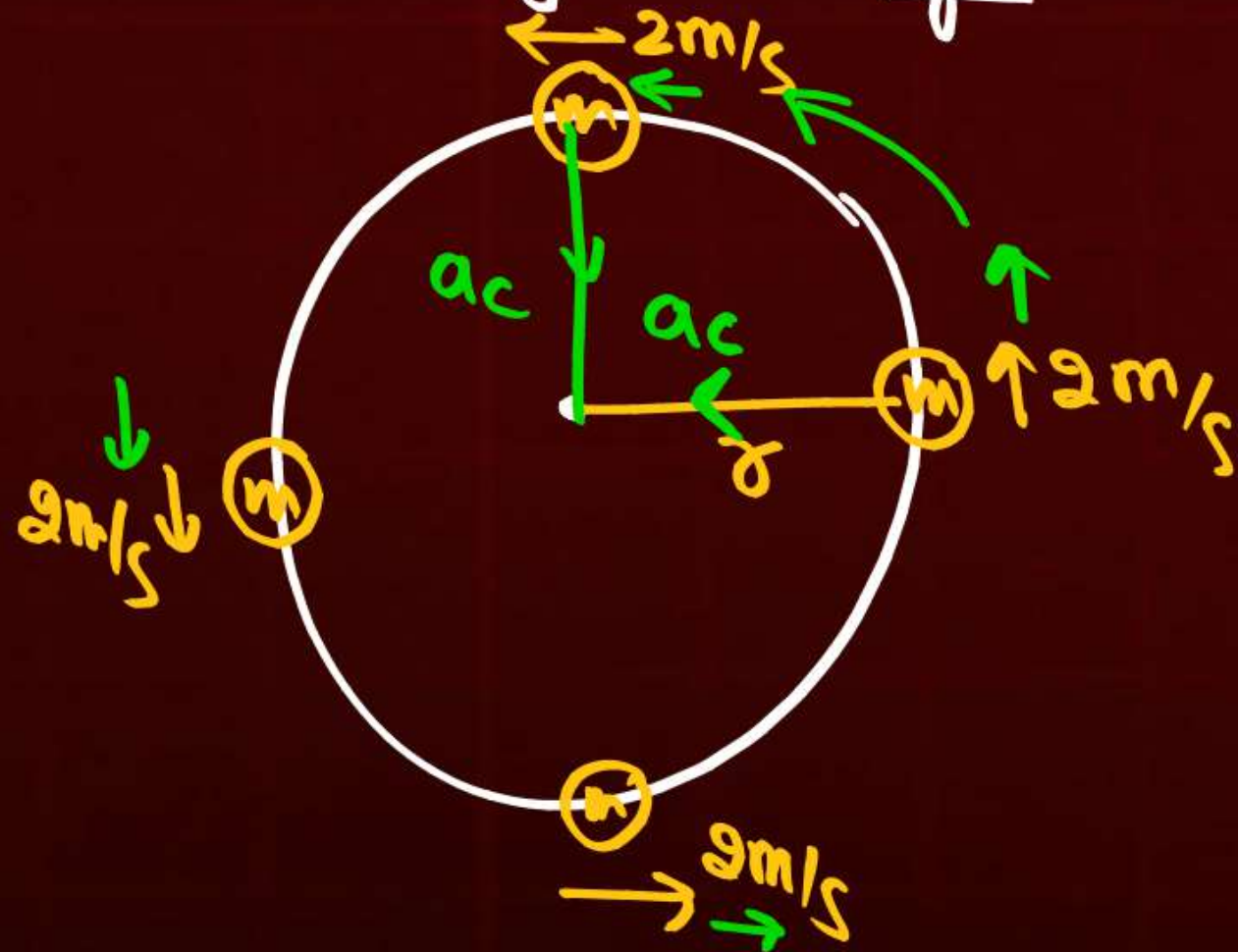
(Centripetal Acceleration)

uniform circular motion

✓ Speed = constant = 2 m/s

✓ Velocity = $\frac{m \times D}{2\text{ m/s}}$

Velocity → change



lecture no. 3

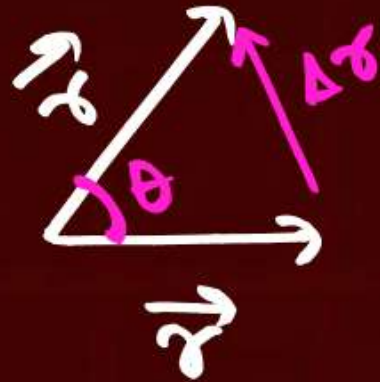
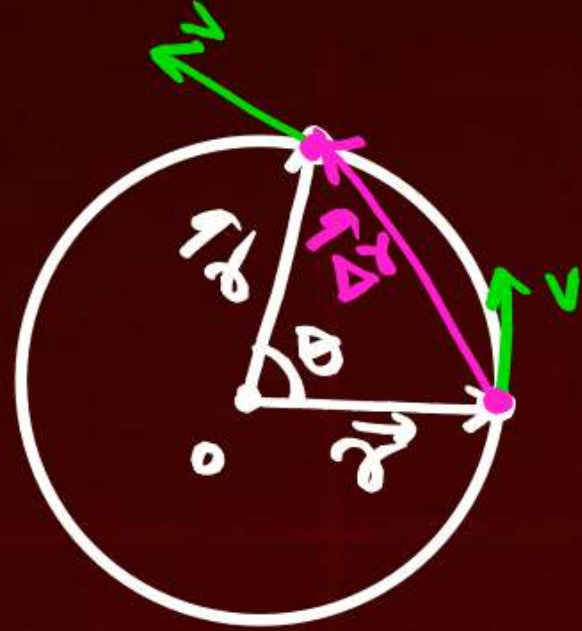
Centripetal acceleration Always act towards the centre

It change the direction of velocity

$$a_c = \frac{v^2}{r} \quad \text{Formula}$$

✓ Derivation

10th → Triangle → Δ Similar Rule
Side Ratio equal



1 photo → 2 Δ

For Small Angle



then this two Δ are similar

then Both Δ have equal Side Ratio

$$\frac{dr}{r} = \frac{dv}{v}$$

Divide b/s by dt

$$\frac{dr}{r dt} = \frac{dv}{v dt}$$

$$\frac{v}{r} = \frac{a}{v}$$

$$\frac{v \times v}{r} = a_c$$

$$\frac{v^2}{r} = a_c$$

$$a_c = \frac{v^2}{r}$$

$$a_c = \frac{v^2}{R}$$

Centripetal Force

the force which cause
the centripetal acceleration

$$a_c = \frac{v^2}{r}$$

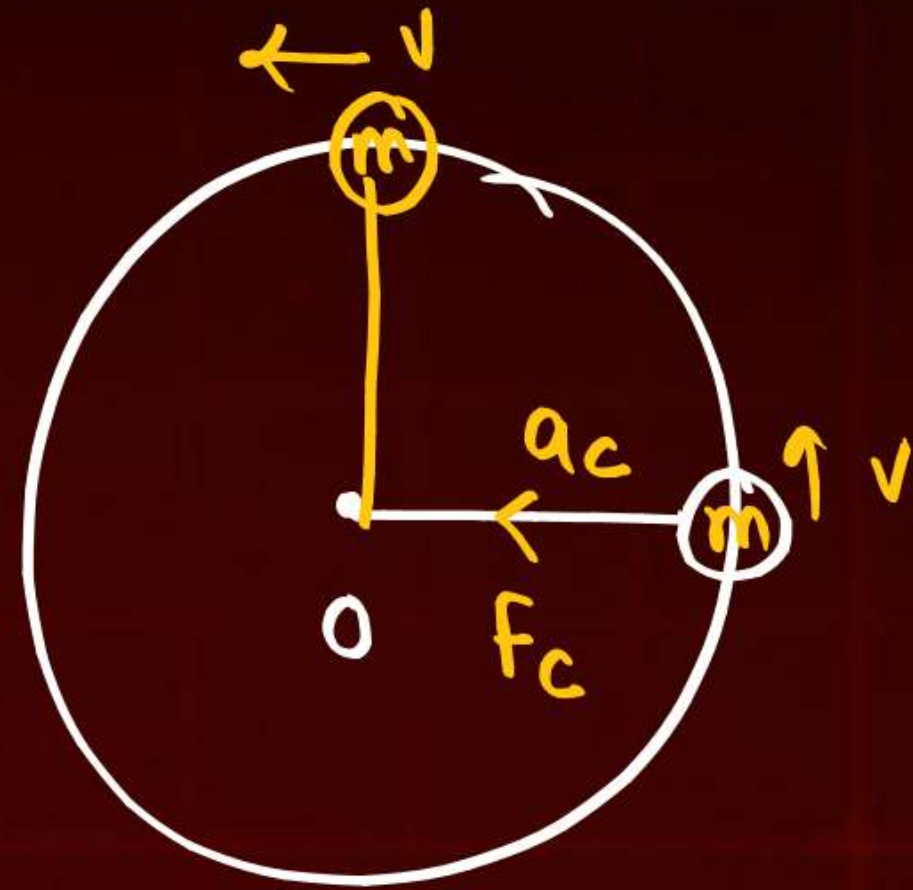
$$F = ma$$

$$F_c = ma_c$$

acts
towards
the centre

$$F_c = m \frac{v^2}{r}$$

acts
towards
the
centre

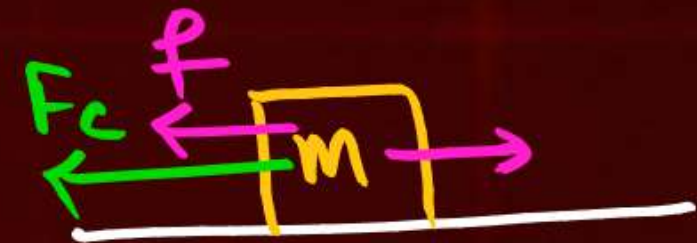
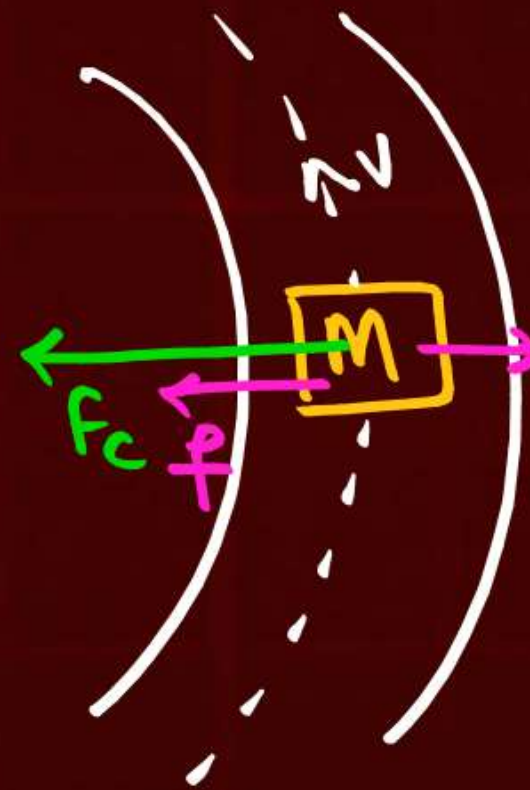
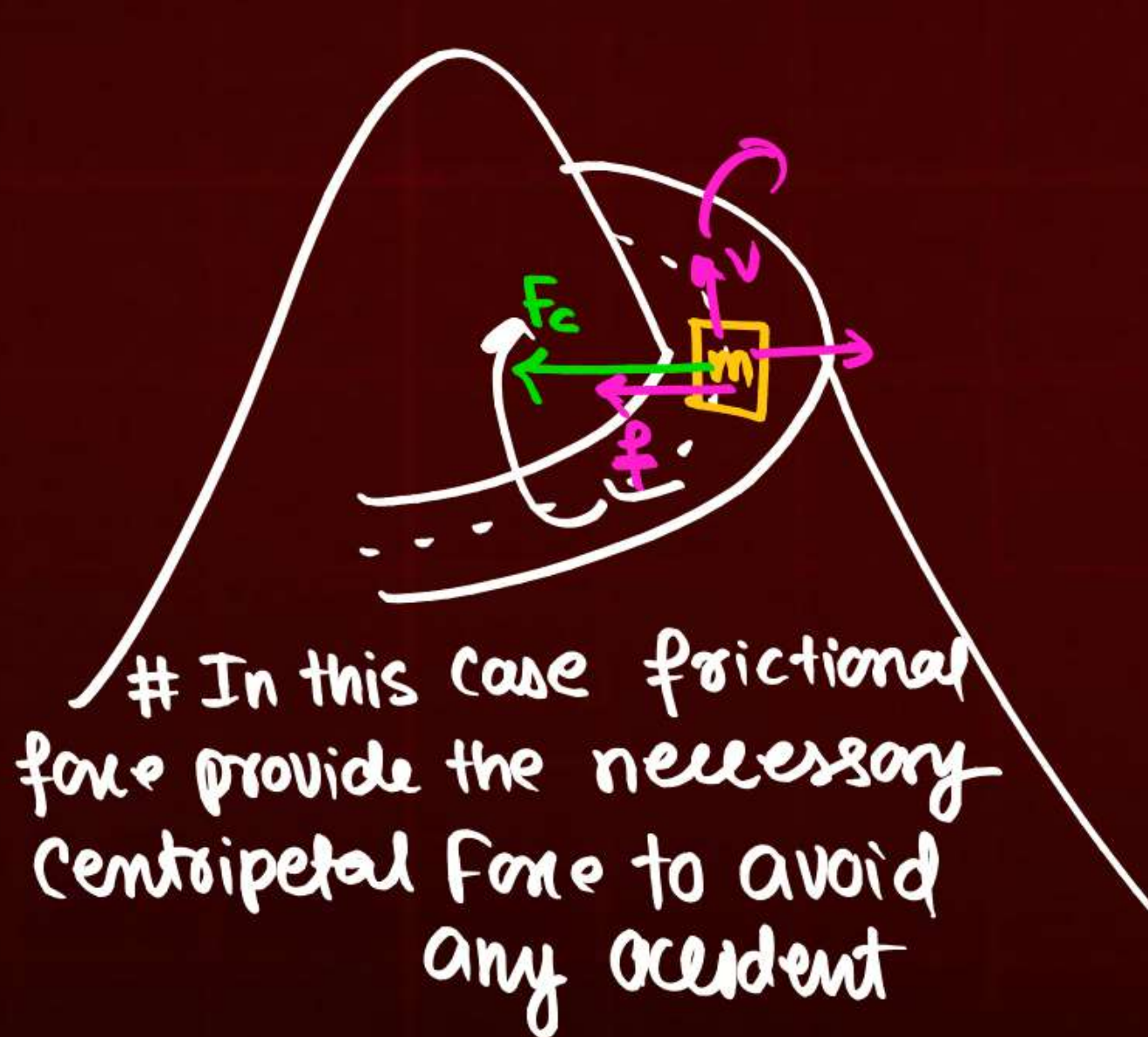


② Maximum safe velocity on a frictional road

on Turning

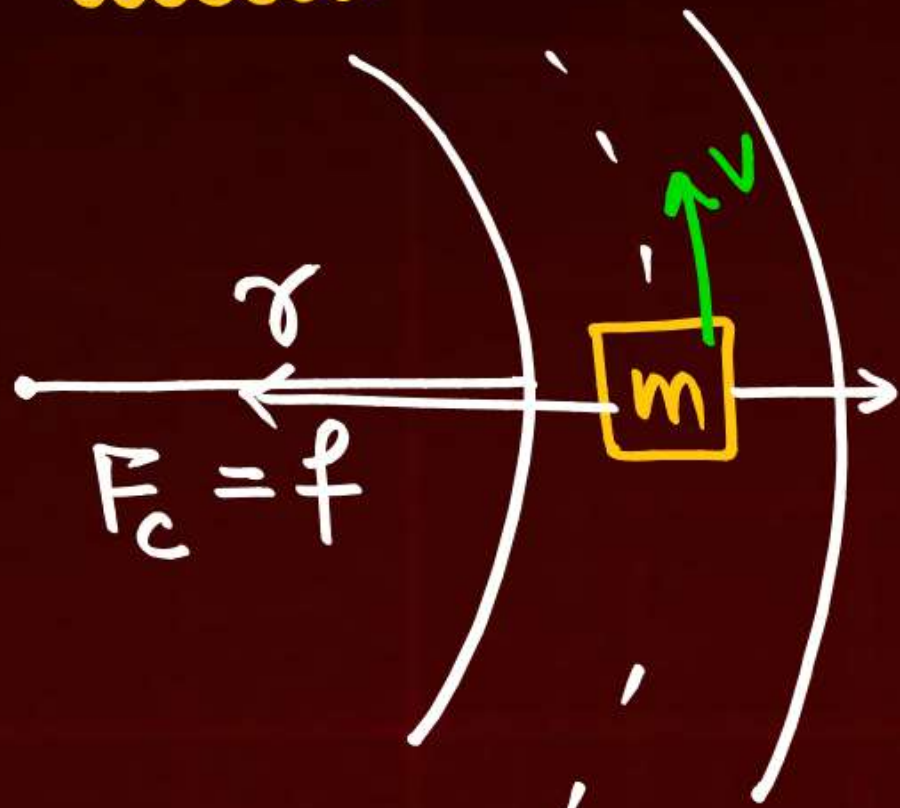


When a vehicle moves along a curved road, it needs a force to keep it turning instead of moving straight. This force is called centripetal force, and on a normal (unbanked) road, it is provided mainly by friction between the tires and the road.

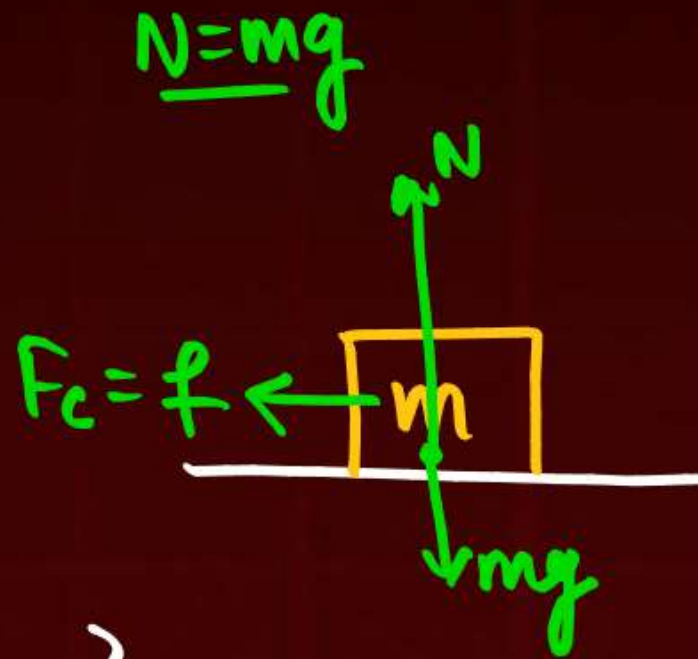


$$V = \sqrt{\mu r g}$$

Derivation



$$\begin{aligned} \# \underline{F_c} &= \underline{f} \\ m \underline{\frac{v^2}{r}} &= \mu \underline{N} \quad \text{mg} \\ \cancel{m} \underline{\frac{v^2}{r}} &= \mu \cancel{m} g \end{aligned}$$



$$\frac{v^2}{r} = \mu g$$

$$v^2 = \mu g r$$

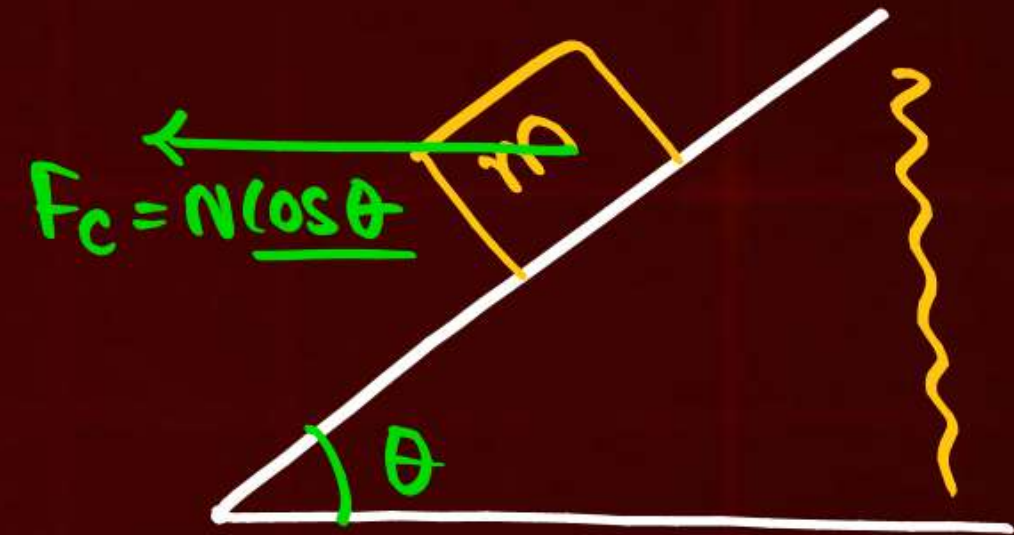
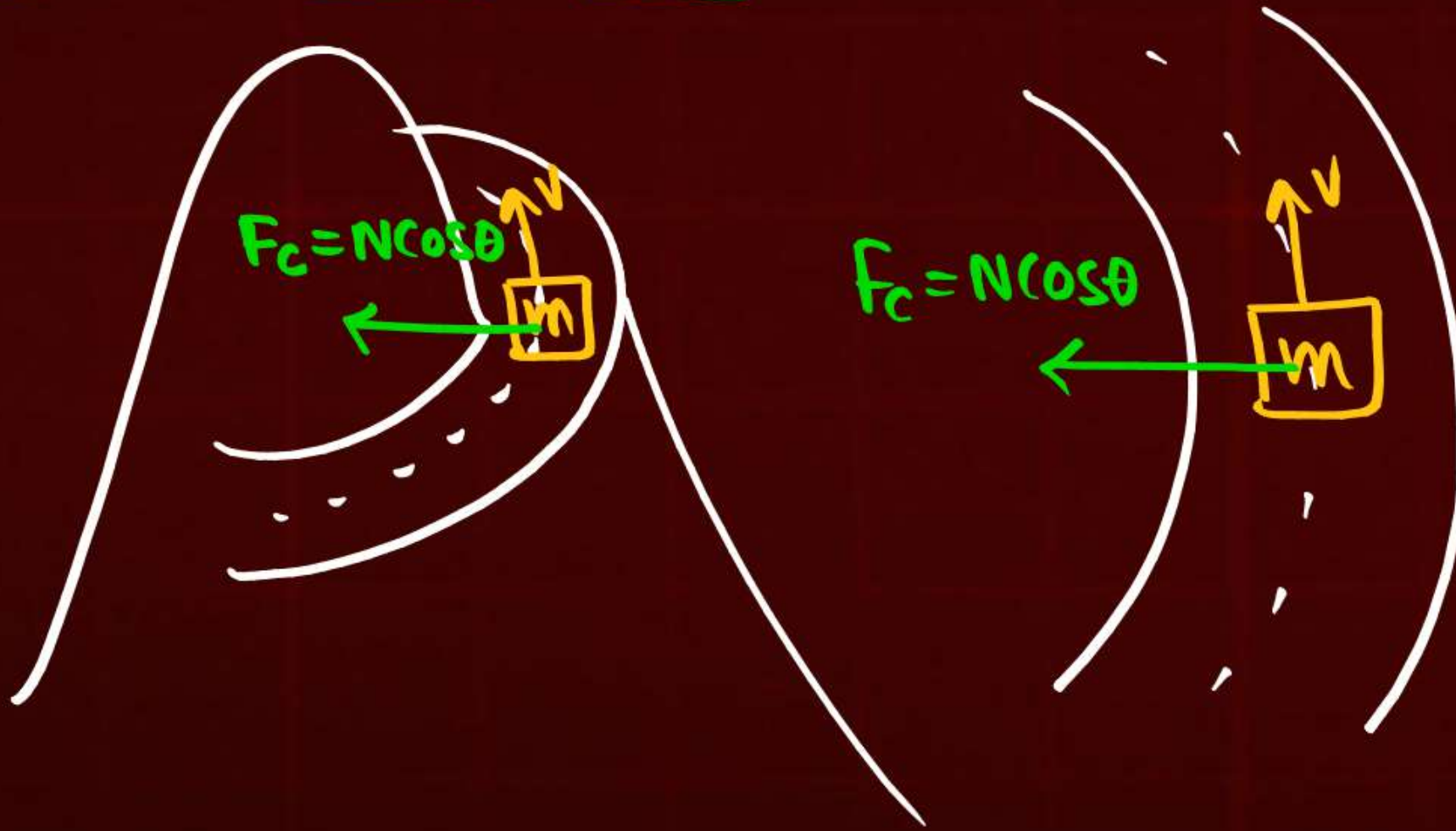
$$v = \sqrt{\mu g r}$$

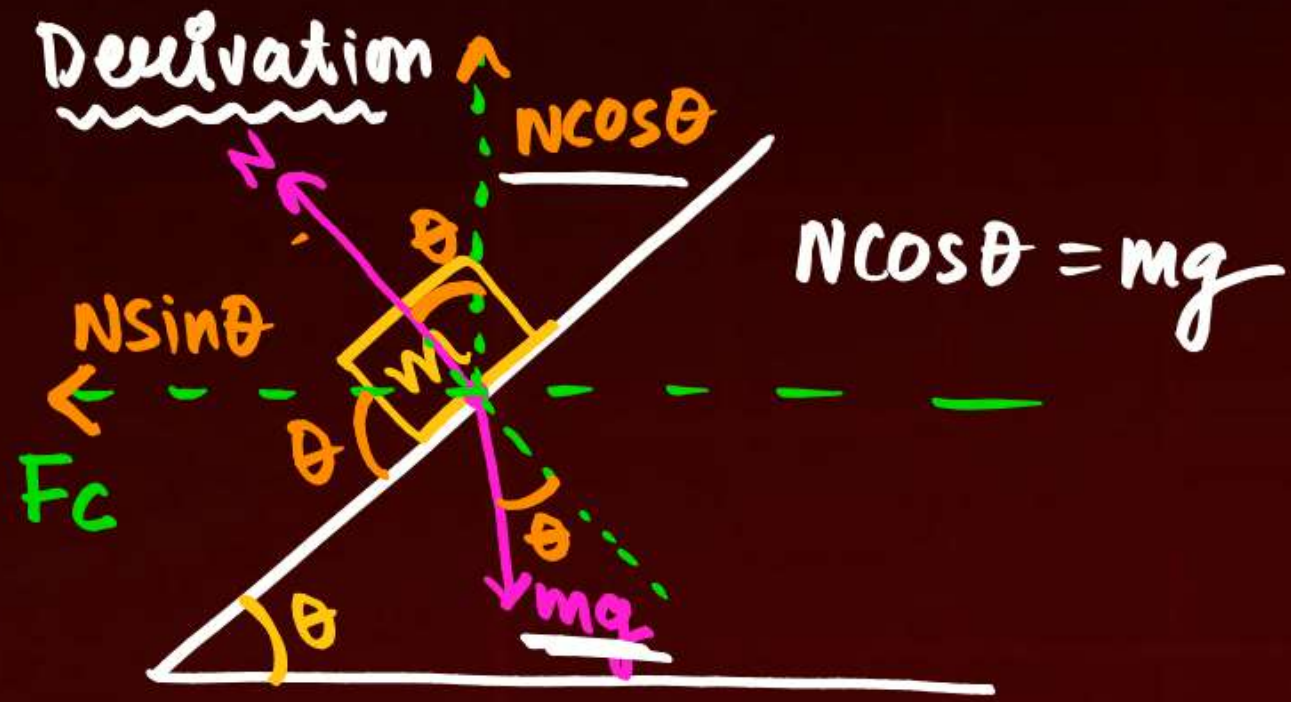
$$v = \sqrt{r g \mu} \quad \checkmark$$

③ Maximum safe velocity on banked road

(Turning) No friction.

When a road is banked (tilted at an angle), the vehicle can take a turn more safely because a part of the normal force provides the required centripetal force-so it doesn't rely only on friction.





$N \sin \theta = F_c$
 $N \sin \theta = \frac{mv^2}{r}$ — (I)

$N \cos \theta = mg$ — (II)

Divide (I) and (II)

$$\frac{N \sin \theta}{N \cos \theta} = \frac{\frac{mv^2}{r}}{mg}$$

$$\tan \theta = \frac{mv^2}{mg r}$$

$$\tan \theta = \frac{v^2}{rg}$$

$$rg \tan \theta = v^2$$

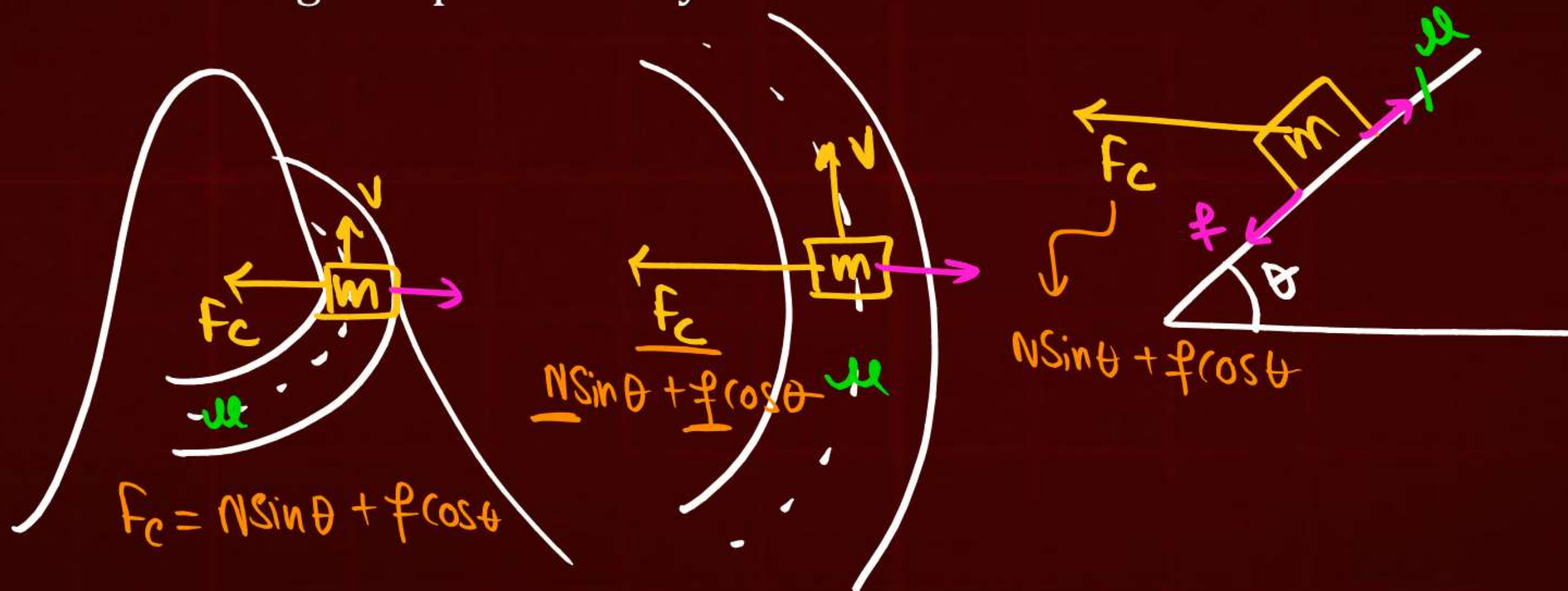
$\sqrt{rg \tan \theta} = v$ Numerical

$$v = \sqrt{rg \mu + rg \tan \theta}$$

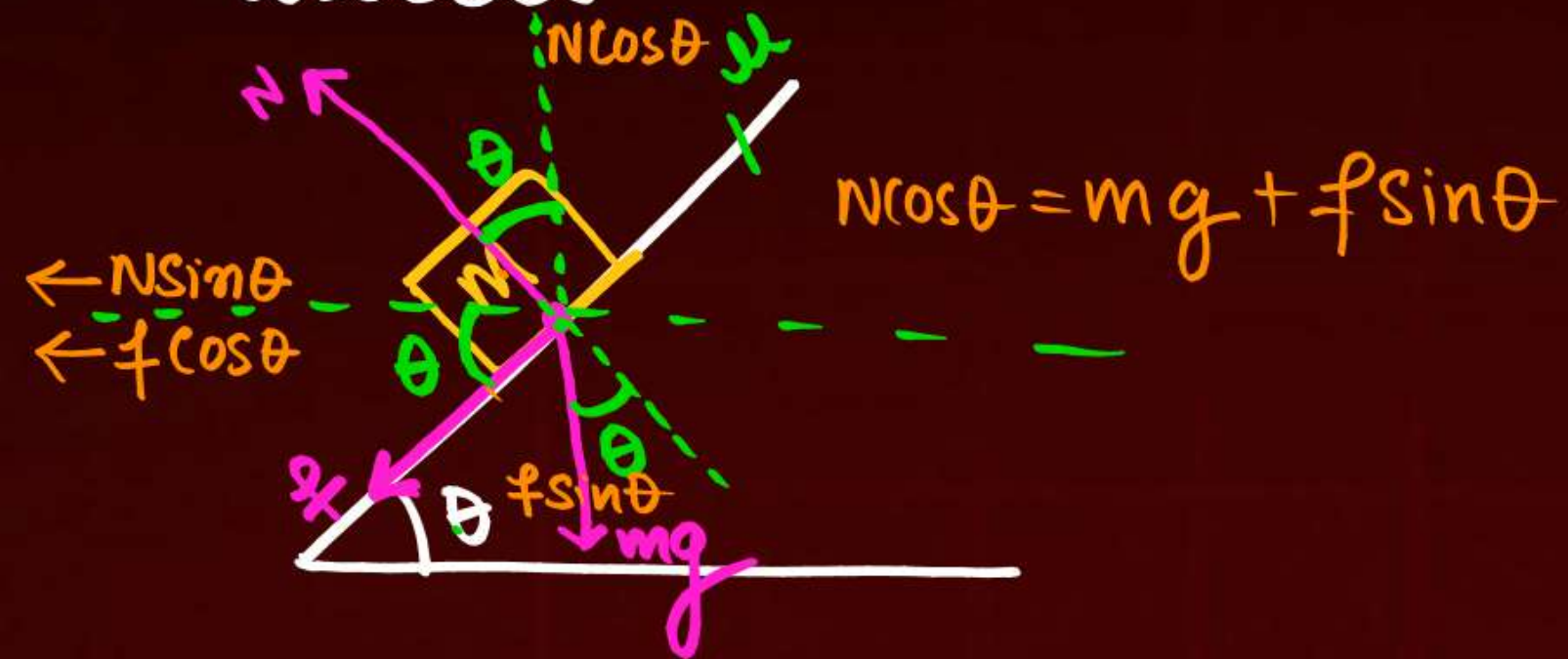
Banked

④ Maximum Safe Velocity on a Banked & Frictional Road

When a road is banked and friction is present, both the normal reaction and friction help provide the centripetal force needed for circular motion. This allows vehicles to take turns at higher speeds safely. Turning ✓



Derivation.



$$\# N \sin \theta + f \cos \theta = F_c$$

$$N \sin \theta + f \cos \theta = \frac{mv^2}{r} \quad \text{--- (I)}$$

$$N \cos \theta = mg + f \sin \theta$$

$$N \cos \theta - f \sin \theta = mg \quad \text{--- (II)}$$

Divide (I) and (II)

$$N \sin \theta + \cancel{f \cos \theta}^{\mu N} = \frac{mv^2}{r} \quad f = \mu N$$

$$N \cos \theta - \cancel{f \sin \theta}^{\mu N} = mg$$

$$\frac{N \sin \theta + \mu N \cos \theta}{N \cos \theta - \mu N \sin \theta} = \frac{\cancel{r} \cancel{m} v^2}{\cancel{r} \cancel{m} g}$$

$$\frac{\cancel{N} (\sin \theta + \mu \cos \theta)}{\cancel{N} (\cos \theta - \mu \sin \theta)} = \frac{v^2}{r g}$$

$$\frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta} = \frac{v^2}{r g}$$

$$\frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta} = \frac{v^2}{rg}$$

$$\frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta} = \frac{v^2}{rg}$$

$$\frac{\tan \theta \cos \theta + \frac{\sin \theta}{\cos \theta} + \frac{\mu \cos \theta}{\cos \theta}}{1 - \frac{\cos \theta}{\cos \theta} - \frac{\mu \sin \theta}{\cos \theta} \tan \theta} = \frac{v^2}{rg}$$

$$\frac{\tan \theta + \mu}{1 - \mu \tan \theta} = \frac{v^2}{rg}$$

$$rg \left[\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right] = v^2$$

$$rg \left[\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right] = v$$

$$v = \left[rg \left[\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right] \right]$$

More Discuss Banked + Friction



$$V = \sqrt{rg \left[\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right]}$$

only friction

$\mu \checkmark$



$$V = \sqrt{rg \left[\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right]}$$

$$V = \sqrt{rg \left(\frac{0 + \mu}{1 - 0} \right)}$$

$$V = \sqrt{rg \times \frac{\mu}{1}} \quad V = \sqrt{rg \mu}$$

only Banking

(Banking)

$\mu = 0$
 $\tan \theta \checkmark$

$$V = \sqrt{rg \left[\frac{\tan \theta + \cancel{\mu}}{1 - \cancel{\mu} \tan \theta} \right]}$$

$$V = \sqrt{rg \left[\frac{\tan \theta + 0}{1 - 0} \right]}$$

$$V = \sqrt{rg \frac{\tan \theta}{1}}$$

$$V = \sqrt{rg \tan \theta}$$

SUMMARY



- $a_c = \frac{v^2}{r}$

$$F_c = m \frac{v^2}{r}$$

- $v = \sqrt{rg\mu}$

- $v = \sqrt{rg \tan \theta}$

- $v = \sqrt{rg \left[\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right]}$

HOMework



Four derivation
cp ✓

Thank
You